

RESEARCH ARTICLE

Establishment of Restoration Trajectories for Upland Tundra Communities on Diamond Mine Wastes in the Canadian Arctic

M. Anne Naeth^{1,2} and Sarah R. Wilkinson¹

Abstract

Mining in the arctic amplifies restoration challenges due to inherent environmental conditions by removing soil, vegetation, and the propagule bank, adding coarse textured wastes with low water holding capacity and nutrients, and introducing salt and metal contamination. Short-term reclamation focuses on rebuilding soil and providing rapid native plant cover for erosion control, supporting longer term reestablishment of ecological processes for sustainable tundra communities that provide essential wildlife habitat. This study evaluated methods to restore soil and plant communities 5 years after implementation of treatments at a diamond mine in the Canadian arctic. Five substrates including mine waste materials (processed kimberlite, glacial till, gravel, and mixes), four amendments (inorganic fertilizer, salvaged soil, sewage sludge, and water treatment sludge), five native species seed mixes and natural recovery were investigated.

Soil and plant response were assessed annually. Soil chemistry was ameliorated with time. Chromium, cobalt, and nickel concentrations in processed kimberlite remained high and potentially toxic to plants. Adding fine textured materials such as glacial till to mine wastes improved nutrient and water retention, which in turn enhanced revegetation. Sewage and inorganic fertilizer increased available nitrogen and phosphorus, plant density and cover. Soil amendment increased species richness. Seeding was essential to establish a vegetation cover. After 5 years, seed mix composition and diversity had no effect on plant community development; soil and plant community properties among treatments changed considerably, providing evidence that restoration in the arctic is dynamic yet slow and success cannot be determined in the short term.

Key words: metals, processed kimberlite, reclamation, seeding, sewage, soil amendment.

Introduction

The arctic's cool, short growing seasons, low precipitation, high winds, and permafrost create harsh environments. Temperatures are above zero only 2–3 months, and frost and snow occur every month. Succession is slow, requiring decades to hundreds of years (Cargill & Chapin 1987; Harper & Kershaw 1997; Forbes et al. 2001). Plant species richness is low, limiting colonizing pools (Forbes & Jefferies 1999). Low temperatures decrease decomposition, nutrient cycling, seed production, and colonization (Haag & Bliss 1973; Billings 1987). Photosynthesis is limited by snow melt, light, and low shoot biomass; poor drainage and aeration due to permafrost limit plant growth and reproduction (Chapin 1983).

Tundra plants are adapted to these conditions. However, industrial activities, such as mining, increase stress, by

removing soil, vegetation, and plant propagules and altering hydrology. Coarse textured mine wastes such as till and crushed rock reduce soil water and nutrient retention capacities and can increase pH and permafrost depth (Johnson 1987; Truett & Kertell 1992). These challenges must be addressed to restore tundra to support local wildlife and indigenous communities.

Primary succession follows severe disturbance due to soil and vegetation loss (Turner et al. 1998; Prach & Pyšek 2001). The inherently harsh environment combined with severe disturbance makes restoration unpredictable and expectations for single trajectories and timelines need reevaluation. While the disturbance regime cannot be altered, stressors such as low nutrients, organic matter, and soil water can be addressed to facilitate colonization by tolerant species. Little arctic research has been conducted on restoration after severe disturbances or in xeric ecosites; most focused on secondary succession in mesic to wet ecosites (Vavrek et al. 1999; Kidd et al. 2006).

By initially addressing stressors such as soil physical and chemical properties, tolerant plants may establish, and with time abiotic changes can facilitate establishment of diverse species (Bradshaw 1997; Bruno et al. 2003). Research in other regions shows adding fine textured soil to coarse mine tailings increases water and nutrient retention and fertilizer provides

¹Department of Renewable Resources, 751 General Services Building, Faculty of Agricultural, Life and Environmental Science, University of Alberta, Edmonton, Alberta T6G 2H1, Canada

²Address correspondence to Dr. M. A. Naeth, email anne.naeth@ualberta.ca

sewage and sludge; only three mixes were sown in fall due to poor weather. In total, there were 162 treatments.

Five 300 m² plots were established in each of three blocks (replicates). Plots were randomly assigned substrates (30 cm over gravel pad), then divided into four randomly assigned amendment strips of equal size. Diavik kimberlite is predominantly silicon, magnesium, and iron (>95% sand). Glacial till is pit overburden material of clay particle to boulder (>1 m diameter) size. Tundra soil (O, A, B horizons), removed during mining and stockpiled, was applied as a 10 cm cover. Substrates had high potassium and sulfur and low phosphorus and nitrogen; therefore 11-52-0 (nitrate-phosphate-potassium) fertilizer was broadcast at 144 kg/ha in summers 2005 and 2006. Untreated sewage (4-cm cover) was applied to 50:50 in September 2004 before fall seeding and to till and 50:50 in June 2005 before spring seeding. Sludge (4-cm cover) from a water treatment plant (particulate from ground rock, lake sediment, open pit till, waste water from seepage and runoff) was applied to PK and 25:75 in June 2005 before spring seeding.

Amendment strips were halved for fall and spring seeding, each half divided into 3–4 m² subplots and randomly assigned a seed mix. Mixes were based on ecosites and competitive abilities (Table 1). Grass mixes were sown to obtain 50 plants/m, with equal species, to reduce erosion and facilitate natural colonization. Grass germination and establishment data from previous research (unpublished) were used to calculate density (220 seeds/m). Forb seed was sown at 13 seeds/m. Plots were seeded in fall (mid-September 2004) and spring (late-June 2005).

Substrate and Vegetation Measurements

Three random soil samples were collected from each reclamation treatment plot in September 2004 at 0–10 and 20–30 cm. Where amendments were added in spring (sewage, sludge) or summer (fertilizer) 2005, sampling followed amendment. Six reference samples were collected in spring 2005 at 0–10 cm (maximum profile) from undisturbed tundra approximately 7 km from active mining. Resampling occurred in summer 2009.

Soil samples were analyzed at a commercial laboratory according to Carter (1993) unless noted. Available phosphate and potassium were determined by modified Kelowna extraction (Qian et al. 1994); available nitrate and sulfate by calcium chloride extraction; sodium adsorption ratio and total cations by saturated paste; pH and electrical conductivity (EC) by meter in saturated paste; cation exchange capacity and extractable cations by barium chloride extraction; total carbon by Leco furnace combustion; inorganic carbon by acid digestion; organic carbon by subtraction; particle size by hydrometer. One third of samples were analyzed for metals restricted in concentration by Canadian Council of Ministers of the Environment (CCME) (silver, arsenic, barium, beryllium, cadmium, cobalt, chromium, copper, molybdenum, nickel, lead, strontium, selenium, tin, thallium, uranium, vanadium, zinc) using acid digestion and inductively coupled plasma mass spectroscopy (SW 846 – 3051/6020); mercury by cold vapour atomic adsorption spectrometry (EPA 200.2/245.1). Water and temperature data in each plot were collected hourly each year (2005–2009) with

HOBO micro stations (Onset Computer Corporation) above the gravel pad and substrate interface (approximately 30 cm) and below surface (approximately 5 cm).

Vegetation was assessed the last week in July or first week in August (2005–2009). Three 20 × 50 cm quadrats were randomly located in each seed mix subplot. Plant density was counted and cover of vegetation, bare ground and litter (dead surface matter) ocularly estimated. Evidence of wildlife grazing (chewed vegetation) and presence of seed heads or flowers were recorded by species in 2007–2009.

Data Analyses

Data were tested for normality and homogeneity of variance. Owing to high heterogeneity of variance, Schrier-Ray-Hare nonparametric two way analysis of variance was performed to assess soil and plant response among treatments (Sokal & Rohlf 1995). Reclamation treatment and seed mix were fixed factors. Significant data were analyzed using Mann Whitney *U* post-hoc tests. Paired *t* tests were conducted to assess plant response between spring and fall seeding and years (substrate 2004, 2009; vegetation 2006, 2009). Categorical response variables (flowering, grazing) were analyzed with Pearson's chi-square test, meeting assumptions for minimum observed values by testing separately for substrate and mix. Low abundance, naturally colonizing species, shrub, moss, and lichen data were converted to categorical presence/absence. Vascular plant species richness was calculated per subplot. Simpson's Diversity Index (*D*) was calculated as sum of species proportions squared (Magurran 2004), from which community evenness ($E = 1 - D$) was calculated. Few plants established in 2005 and 2006 precluding some analyses. SPSS 18 and a confidence level of 95% were used for all tests.

Results

Effect of Reclamation Treatment on Soil

Soil property trends for both depths were similar unless noted, thus 0–10 cm data are provided. Substrates and reference soil were loamy sand to sandy loam texture (data not shown). By year 5, reclamation treatment pH was higher than reference soil; highest in PK then PK mixes, gravel and till (Table 2). PK, PK mix, and gravel pH were moderated by soil and sewage. Unamended PK and PK mix had significantly higher pH than gravel or till, the latter most similar to reference soil. EC was generally <2 dS/m and sodium adsorption ratio <2 and not a concern for plants. Both were higher in treatments than reference soil, with the exception of gravel. Within 5 years, pH increased in PK, significantly in PK-unamended and PK-fertilizer, and decreased in till, significantly with sewage. EC in most treatments decreased from years 1 to 5, significantly in till-sewage, PK-unamended, PK-fertilizer, PK-sludge, and 50:50-unamended. EC significantly increased in till-unamended and till-fertilizer to values potentially unacceptable for plants. Treatments exceeding this initially (till-sewage, PK-unamended, PK-fertilizer) were similar to others by year 5.

Table 2. Soil properties at 0–10 cm depth in reclamation treatments 5 years following seeding.

Substrates	pH	EC dS/m	TOC %	NO ₃ mg/kg	PO ₄ mg/kg	K mg/kg	SO ₄ mg/kg	CEC Meq/100 g	Cr	Co	Ni	VWC %	Temp °C
Gravel	7.1 ^a (0.3)	0.2 (0.0)	0.11 ^a (0.02)	1.0 (0.0)	3.8 (0.8)	48.8 (6.8)	1.9 (0.3)	1.7 ^a (0.1)	18.7 (10.3)	4.1 (1.8)	42.7 (28.6)	3.8 (0.6)	13.5 (1.4)
Gravel fertilizer	7.2 ^a (0.1)	0.2 (0.0)	0.13 ^a (0.03)	1.0 (0.0)	18.0 (5.1)	46.4 (7.1)	1.7 (0.2)	1.7 ^a (0.2)	77.3 (26.4)	17.3 (7.1)	276.5 (131.1)	3.1 (0.6)	14.0 (1.2)
Gravel soil	6.3 ^b (0.2)	0.3 (0.1)	0.80 ^b (0.3)	2.3 (0.4)	7.8 (1.4)	53.8 (10.8)	4.4 (1.2)	3.7 ^b (1.0)	27.8 (6.9)	4.3 (0.5)	29.5 (9.0)	3.6 (0.5)	14.2 (1.2)
PK	8.5 ^a (0.0)	0.6 (0.0)	0.26 ^a (0.02)	1.0 (0.0)	2.5 (0.2)	265.4 ^a (28.2)	13.6 (1.6)	8.2 (1.1)	339.0 ^a (13.9)	77.7 ^a (2.3)	1606.7 ^a (143.3)	10.4 (0.5)	15.4 (1.3)
PK fertilizer	8.5 ^a (0.0)	0.7 (0.1)	0.53 ^a (0.02)	1.0 (0.0)	7.4 (1.4)	259.2 ^{ab} (13.1)	13.8 (2.4)	7.5 (0.6)	334.0 ^a (22.5)	70.4 ^a (4.1)	1453.3 ^a (184.1)	7.7 (0.3)	15.0 (1.3)
PK soil	8.0 ^b (0.2)	1.2 (0.3)	0.75 ^b (0.30)	5.1 (2.8)	5.1 (1.7)	203.5 ^b (19.9)	30.3 (7.7)	8.5 (0.6)	228.0 ^b (29.0)	47.2 ^b (9.2)	832.0 ^b (169.0)	6.5 (0.9)	14.9 (1.3)
PK sludge	8.4 ^a (0.1)	0.9 (0.1)	0.23 ^a (0.03)	5.5 (2.8)	2.8 (0.3)	282.6 ^a (16.3)	18.7 (1.5)	8.4 (0.7)	343.3 ^a (127.1)	69.9 ^a (3.4)	1333.3 ^a (127.1)	10.0 (0.6)	15.4 (1.3)
Till	5.7 ^a (0.3)	3.2 (1.0)	0.25 ^a (0.04)	55.7 (25.0)	7.3 ^a (0.6)	83.6 ^{ab} (11.6)	56.8 (14.7)	2.7 ^a (0.5)	43.9 (2.8)	8.9 (0.4)	40.7 (12.9)	5.7 (1.1)	14.6 (1.5)
Till fertilizer	5.9 ^{ab} (0.3)	4.1 (0.7)	0.19 ^a (0.01)	94.5 (23.9)	14.3 ^a (4.3)	92.8 ^{ab} (10.6)	66.5 (14.7)	3.8 ^a (0.3)	44.9 (3.7)	9.2 (1.2)	42.7 (18.0)	13.7 (0.9)	14.6 (1.2)
Till soil	5.9 ^{ab} (0.2)	1.4 (0.2)	0.84 ^b (0.24)	14.5 (3.7)	7.7 ^a (0.8)	81.6 ^b (4.1)	65.7 (32.3)	4.4 ^b (1.1)	37.4 (2.0)	6.9 (0.2)	52.5 (5.6)	6.3 (1.1)	14.2 (1.2)
Till sewage	6.0 ^b (0.2)	0.9 (0.3)	0.63 ^b (0.22)	16.9 (8.7)	197.6 ^b (89.9)	105.5 ^a (24.0)	15.9 (3.2)	3.6 ^{bc} (0.8)	60.3 (2.4)	10.6 (1.0)	91.1 (19.3)	16.2 (1.1)	n/a
50:50	8.2 ^a (0.0)	0.5 (0.0)	0.24 ^a (0.01)	1.0 (0.0)	4.95 ^a (1.1)	156.4 ^a (10.6)	8.0 (0.8)	3.7 ^a (0.5)	198.3 ^{ab} (28.3)	44.9 ^{ab} (7.2)	731.7 ^{ab} (139.0)	n/a	14.2 (0.9)
50:50 fertilizer	8.3 ^a (0.1)	1.3 (0.25)	0.23 ^a (0.01)	7.0 (2.9)	9.6 ^a (1.9)	190.1 ^b (19.5)	38.1 (10.1)	4.4 ^a (0.5)	202.7 ^{ab} (7.5)	50.7 ^b (2.3)	837.0 ^b (36.4)	7.8 (0.6)	13.7 (1.1)
50:50 soil	7.9 ^b (0.2)	1.0 (0.1)	0.59 ^b (0.20)	3.7 (0.9)	6.7 ^a (1.0)	154.6 ^a (8.4)	25.2 (3.1)	5.5 ^a (0.5)	153.9 ^a (50.5)	31.8 ^a (6.8)	518.3 ^a (130.4)	7.8 (1.0)	14.4 (1.2)
50:50 sewage	7.7 ^b (0.2)	1.3 (0.2)	0.97 ^c (0.31)	21.5 (7.8)	159.0 ^b (76.4)	193.3 ^{ab} (22.9)	14.6 (1.8)	7.1 ^b (0.7)	219.7 ^b (28.3)	48.9 ^b (4.1)	846.7 ^b (93.1)	7.1 (0.7)	13.3 (1.0)
25:75	8.0 ^a (0.1)	1.4 (0.3)	0.21 ^a (0.05)	5.5 (1.6)	5.6 (0.6)	141.7 (17.8)	47.7 (10.5)	3.8 ^a (0.4)	155.0 ^a (25.6)	35.5 ^a (5.1)	561.0 ^a (103.8)	12.8 (0.5)	14.8 (1.3)
25:75 fertilizer	8.2 ^a (0.0)	1.0 (0.1)	0.23 ^a (0.04)	2.1 (0.8)	11.9 (3.7)	166.5 (11.3)	29.4 (2.5)	4.6 ^a (0.4)	153.4 ^a (55.6)	32.7 ^a (6.2)	510.0 ^a (134.0)	10.3 (1.2)	14.8 (1.3)
25:75 soil	7.3 ^b (0.3)	1.5 (0.3)	0.61 ^b (0.34)	13.4 (7.6)	8.1 (1.34)	137.4 (12.3)	38.4 (9.7)	5.1 ^b (1.0)	93.1 ^b (35.0)	19.7 ^b (10.2)	300.5 ^b (181.5)	5.9 (0.5)	15.0 (1.3)
25:75 sludge	8.0 ^a (0.1)	1.8 (0.4)	0.23 ^a (0.02)	15.4 (6.0)	6.6 (0.7)	161.6 (16.5)	46.0 (10.8)	3.7 ^a (0.5)	161.7 ^a (25.0)	35.7 ^a (7.4)	555.7 ^a (149.8)	13.2 (1.6)	14.5 (1.3)
Reference soil	5.3 (0.2)	0.1 (0.0)	4.02 (2.29)	1.0 (0.0)	5.0 (0.7)	55.8 (12.8)	3.2 (1.8)	9.8 (4.5)	26.0 (1.5)	5.1 (0.3)	18.0 (1.0)	n/a	n/a

PK, processed kimberlite; VWC, volumetric water content. Numbers in parentheses are standard error of means. Nitrate (NO₃), phosphate (PO₄), potassium (K), and sulphate (SO₄) are available nutrients. Mean daily VWC and temperature at soil surface (5 cm) during the months of July and August 2008. Within substrates (gravel, PK, till, 50:50, 25:75), means which do not share a common letter are significantly different at $p = 0.05$.

Organic carbon was lower in treatments than reference soil (Table 2), although differences were statistically insignificant. Available nutrients were higher in treatments than reference soil with gravel treatments most similar. PK in treatments significantly increased available potassium and till increased available sulfate relative to reference soil and gravel. 50:50-sewage and till-fertilizer had significantly greater available nitrate than gravel, PK, 50:50 (except 50:50-sewage), 25:75-unamended, and 25:75-fertilizer. Available nitrate in 25:75-sewage was only greater than unamended gravel, PK and 50:50 and fertilized gravel and PK. At depth, nitrate was significantly greatest in till-fertilizer; sewage had no effect. Cation exchange capacity of PK was similar to reference soil. Nitrate decreased with time, significantly in PK, and increased in till-unamended and till-fertilizer. Sewage had significantly greater available phosphate, higher in year 5 than 1, significantly with fertilizer in PK and gravel. Sulfate significantly decreased with time, except in till-fertilizer, till-unamended, and till-soil.

Metal concentrations were low, changing little with time (Table 2). Year 1 chromium and nickel in all treatments, cobalt in PK and copper and zinc in till-sewage, exceeded CCME guidelines (Drozdowski et al. 2012). Molybdenum and selenium were at or near CCME in sewage. Year 5 chromium, cobalt, and nickel exceeded CCME (64, 50, 50 mg/kg, respectively) in PK treatments. At 0–10 cm, unamended PK consistently had greater than 300 mg/kg chromium and 1,000 mg/kg nickel, with unamended PK mixes lower. High metals in sewage were ameliorated with time. In PK and PK mixes both metals increased with time with soil or sewage amendment. Increases were not found in till or gravel, except gravel-fertilizer where both metals increased at both depths.

Annual precipitation varied (104–203 mm), with 2008 almost twice as high as previous years. Volumetric water content was low in all treatments (Table 2) at 5–10% throughout June to August (Drozdowski et al. 2012). Treatments with >10% in all years were 25:75-sludge, 25:75-unamended, till-fertilizer, and till-sewage. In 2005, 50:50-sewage and PK-unamended had high water relative to other treatments, with <10% from 2006 to 2008. Till-fertilizer and 25:75-unamended were least affected by precipitation with more constant water than other high water treatments. Water content was less at the interface; treatments with highest surface water also had highest at depth. Surface or interface soil temperature differed little among treatments (Table 2) or years. Higher temperatures in PK persisted due to its black color (Drozdowski et al. 2012).

Effect of Reclamation Treatment on Vegetation

Seeding season had no significant effect on plant abundance, thus analyses were conducted on spring data. There was no interaction between reclamation treatment and seed mix.

In years 1–4, treatment had no significant effect on plant abundance. Although statistically insignificant, in all years, plant density and cover were consistently lowest in PK-unamended; by year 3 PK-sludge and PK-fertilizer were similar. Year 5 density ranged from 0 to 180 plants/m, significantly highest in 50:50-sewage, till-sewage, and till-fertilizer

(Fig. 1). Plant cover was <15% the first 5 years (Fig. 2). Year 5 cover was significantly highest in till-sewage and 50:50-sewage. Amendment significantly increased density and cover relative to the unamended control in gravel and till and density in 25:75. Fertilizer or soil increased density and cover relative to the control in PK and sewage in 50:50. Soil PK and 25:75 had significantly greater density and cover than other amendments. Gravel-fertilizer and gravel-soil were similar; within till and 50:50, sewage or fertilizer had best plant response.

Amendments initially improved PK for plant establishment; with time density and cover significantly decreased (Figs. 2 and 3). In year 2, amended gravel and 25:75-sludge had highest densities; by year 5 density and cover significantly declined. Till-fertilizer, a top performer in year 2, had high density and cover through year 5. Positive effects of sewage on density did not occur until year 4. Beyond year 2, amendments other than sewage had little effect on cover which declined except in sewage and till-fertilizer (Fig. 3).

Species richness was significantly affected by treatment in years 3 and 5, with near significant results in year 4. Unamended plots had less than 6 species, gravel-soil had up to 12. PK-unamended, PK-fertilizer, and PK-sludge had significantly fewest species (Table 3). Species richness was significantly greatest in soil; similar in gravel and till to fertilizer. Richness significantly declined in PK-fertilizer and PK-sludge between years. Simpson's Diversity Index was <0.5 for most treatments, indicating low evenness, and did not differ significantly among treatments. Between years, community diversity increased except with sewage or sludge, significantly in till-fertilizer, 25:75-fertilizer, 50:50-fertilizer, 50:50-soil, and till-soil.

Flowering occurred in 30% of quadrats each year. PK had significantly lower flowering than other substrates and 50:50 significantly more, except till ($p=0.000$) (Table 3). Soil significantly increased flowering relative to unamended PK and 25:75 ($p=0.000$, $p=0.006$). Within till and 50:50, flowering was similar among soil and sewage and significantly greater than unamended ($p=0.000$ and $p=0.027$). Flowering did not differ among gravel treatments.

By year 5, litter developed (Table 3). Till-sewage, till-fertilizer, and 50:50-sewage, with highest plant cover, had 2, 4, and 10% litter, respectively; other treatments had <1%. Litter was significantly less in PK ($X^2=76.15$, $p=0.000$). Sludge decreased litter in PK and 25:75, while soil increased it (PK: $X^2=8836$, $p=0.000$ and 25:75: $X^2=22.44$, $p=0.000$). Amendments increased litter relative to till-unamended ($X^2=36.31$, $p=0.000$).

Frequency of naturally colonizing species was significantly affected by substrates except if amended with soil (Table 3). PK and 25:75 had significantly least natural recovery, with one occurrence each ($X^2=78.93$, $p=0.000$). Sewage significantly increased natural recovery in till ($X^2=32.99$, $p=0.000$) and 50:50 ($X^2=54.45$, $p=0.000$). *Epilobium angustifolium* was the dominant natural colonizer. Shrubs had low occurrence, significantly more frequent in gravel (4% of plots) than other treatments ($p=0.000$).

Moss and lichen provided evidence of plant community development in year 5. Moss cover was greatest in till-sewage (4.5%), till-fertilizer (1.7%), 50:50-soil (1.1%), and gravel-fertilizer (0.9%). None established in PK. Amendments significantly increased moss in gravel, 50:50 and 25:75

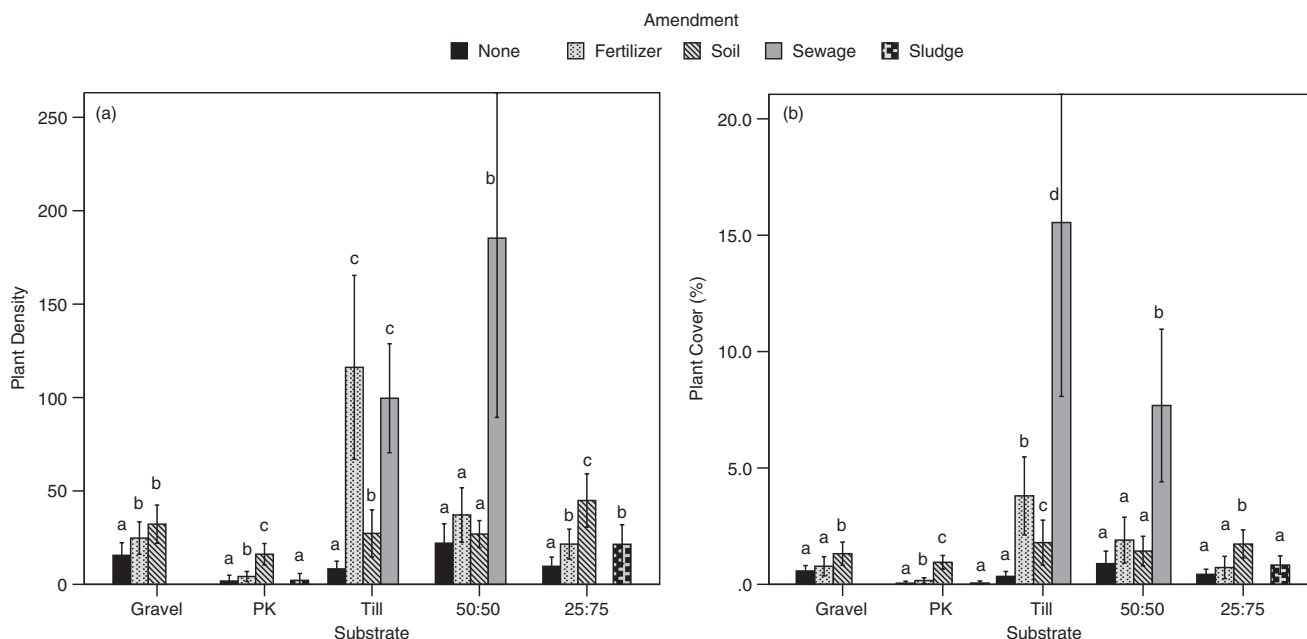


Figure 1. (a) Mean plant density (per m²) and (b) cover in reclamation treatments 5 years following seeding with native grasses and forbs. Error bars represent ± 2 standard errors. Within substrates, columns that do not share a common letter are significantly different. PK, processed kimberlite; 50:50, 50% processed kimberlite and 50% till (by volume); 25:75, 25% processed kimberlite and 75% till.

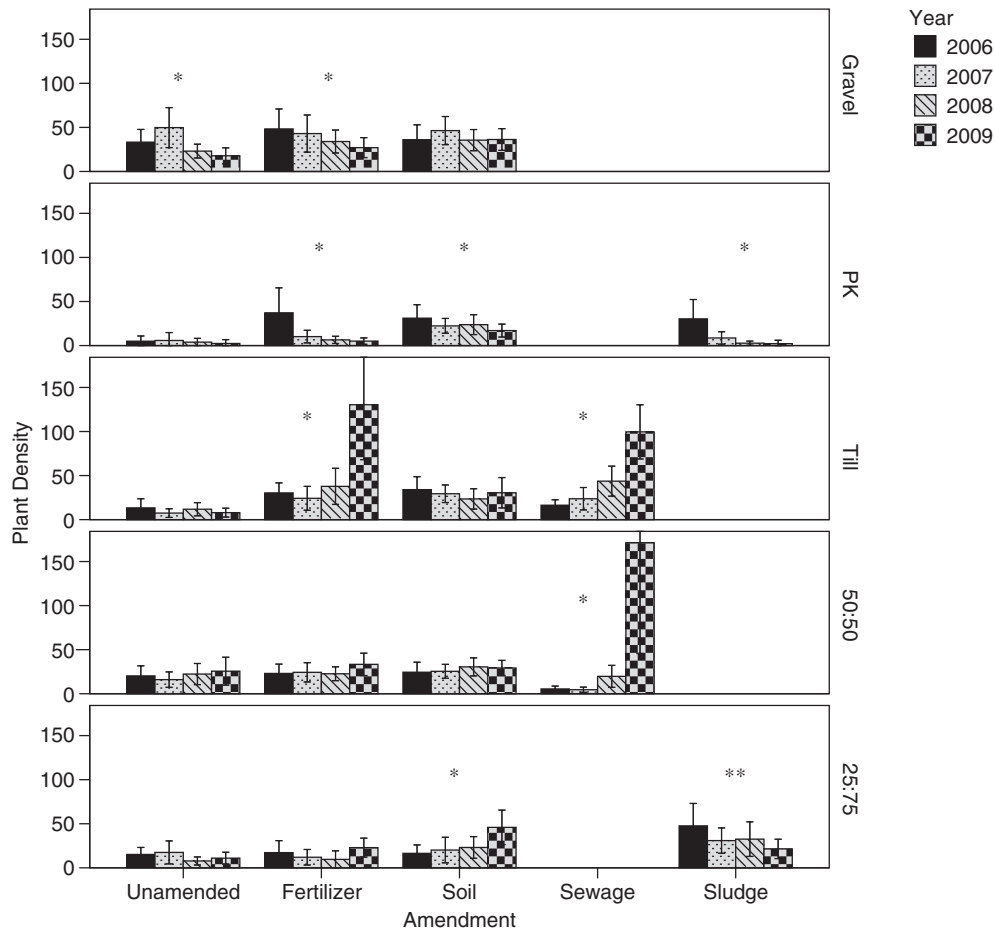


Figure 2. Mean plant density (per m²) in reclamation treatments seeded with native grasses and forbs. Asterisks indicate a significant difference between first and last year of monitoring at * 0.050 or ** 0.100 level. PK, processed kimberlite; 50:50, 50% processed kimberlite and 50% till (by volume); 25:75, 25% processed kimberlite and 75% till.

relative to the control ($p=0.000$, $p=0.000$, $p=0.039$, respectively) and till-sewage had significantly more than other till treatments ($p=0.000$). Lichen established in till-soil and gravel-unamended with negligible cover.

Grazing was evident in 33% of quadrats in year 3, 14% in year 4, and 6% in year 5. In year 3, till was grazed significantly less than other substrates ($p=0.000$) and in year 4 gravel significantly more ($p=0.000$) (Table 3). By year 5, there were no differences among substrates; within substrates, unamended treatments had significantly less grazing than amended.

Effect of Seed Mix on Vegetation

In years 2–4, seed mix significantly affected plant abundance. All seeded species established. Natural recovery and forb mix had significantly fewest plants the first 4 years; by year 5, density was similar to other mixes (Fig. 1). Diverse mix had fewer plants and cover than non-aggressive mix but more species; two potentially competitive species (*Calamagrostis canadensis* and *Festuca rubra*) had low establishment. Drought tolerant mix had greatest density and cover and most *Poa glauca*. *Puccinellia nuttalliana* was most abundant, except in adaptable mix, although

only seeded in diverse mix and non-aggressive mix. Initially, natural recovery and forb-mix had significantly lower richness than grass mixes and natural recovery significantly lower than forb-mix. By year 5, no differences occurred among natural recovery or mixes; richness ranged from 0 to 13 species. Simpson's Diversity Index was less than 0.5 and did not differ significantly among mixes. With time, community diversity increased significantly in revegetation treatments except adaptable-mix. Natural colonizers (excluding species sown in adjacent plots) established in 8% of natural recovery plots, not significantly different than seed mixes. Most plant establishment in natural recovery was due to grass egress from adjacent seeded plots. Natural recovery had significantly lower flowering than seed mixes, with no difference among mixes.

Discussion

Dynamic plant communities established within 5 years on substrates created with diamond mine waste materials. This demonstrates potential for restoration of large and intense arctic disturbances from natural resource extraction. Our research confirms

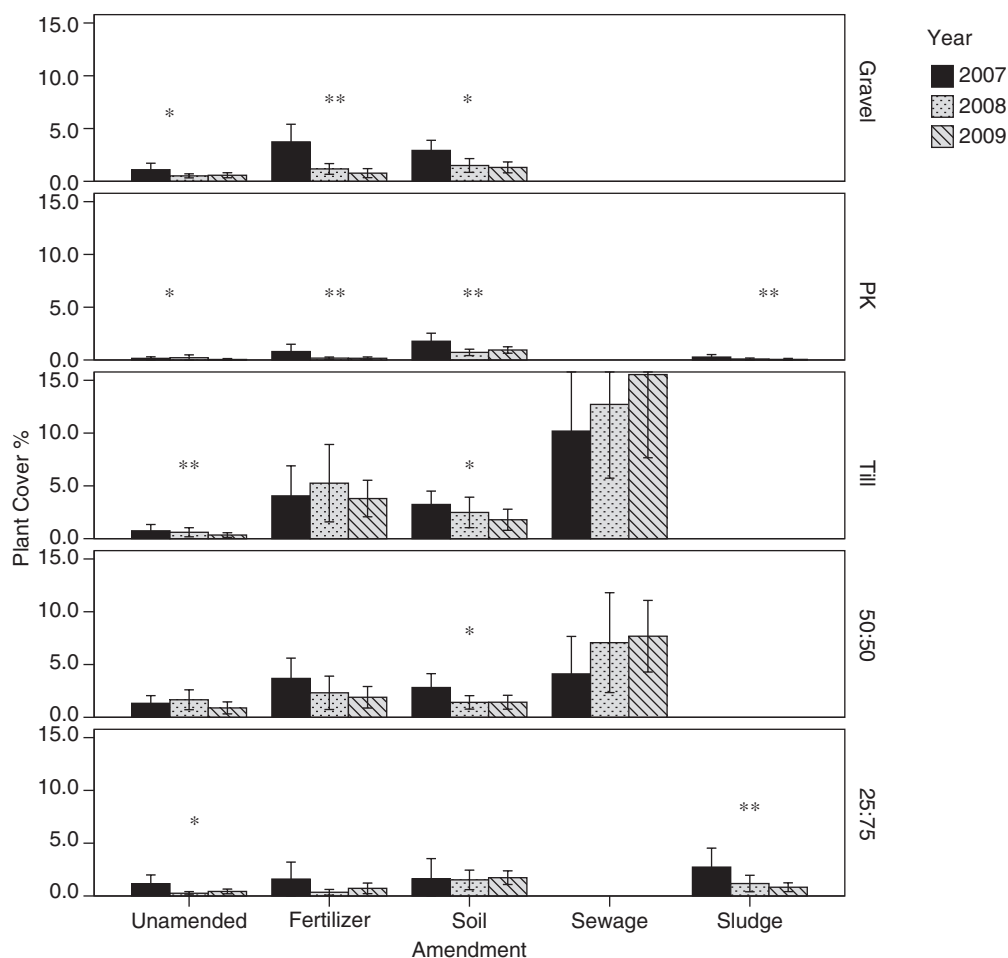


Figure 3. Mean plant cover in reclamation treatments seeded with native grasses and forbs. Asterisks indicate a significant difference between first and last year of monitoring at * 0.050 or ** 0.100 level. PK, processed kimberlite; 50:50, 50% processed kimberlite and 50% till (by volume); 25:75, 25% processed kimberlite and 75% till.

the need for adding fine textured material and organic matter to coarse mine wastes such as processed kimberlite and gravel and seeding native species for sustainable soil and plant community development.

Adding fine textured material and organic matter accelerated soil and plant development, supporting results of others reclaiming highly disturbed sites (Cargill & Chapin 1987; Wong 2003). The exception was treatments primarily composed of PK, where arrested development occurred. Plant community development in PK ceased and time did not ameliorate it. While organic matter ameliorated mine spoil elsewhere (Smyth 1997; Reid & Naeth 2005), capping PK with soil only provided short-term benefits as it eroded. Fertilizer and sludge had little effect, likely due to PK's coarse texture resulting in nutrient leaching. Other mine wastes with severe physical limitations including gravel and tailings sands can sustain adequate plant growth (Li & Fung 1998); therefore PK chemical properties (metals, nutrients) likely limit plant establishment and survival in our study.

The gravel substrate was promising for initial revegetation. In the first years following reclamation, gravel-fertilizer and

gravel-soil exceeded most treatments with greater plant density, species richness, and evenness. Seed readily germinated in gravel, perhaps due to its rough surface increasing microtopographic relief and providing shelter for seed and seedlings (Peterson & Billings 1980). Soil pH and available nutrients were similar to undisturbed soil, a benefit to tundra plants which are extremely sensitive to pH and nutrient fluctuations (Viereck 1966). Establishment subsequently subsided, while increasing in other treatments, resulting in a few species (*Poa glauca*, *Festuca saximontana*) with small plants and suggesting substrate requirements for initial plant establishment are not the same as for sustained growth.

Similar to gravel, dynamic responses occurred in sewage treatments. Sewage initially had very low plant densities, with increased establishment and growth in years 4 and 5 resulting in highest cover. With time metals and salinity decreased in our study, while water retention remained relatively high. With time nutrients bound to organic matter may also have been released. Litter was high and its decomposition adds nutrients and organic matter, building soil structure. High cover and

Table 3. Plant community properties in reclamation treatments 5 years following seeding with native grasses and forbs.

Substrate	Species Richness	Community Evenness	Frequency (%)					
			Natural Recovery	Flowers/Florets	Grazing Evidence	Moss	Litter	Shrubs
Gravel	4.2 ^a (0.3)	0.57 (0.06)	5	28	1	7	47	3
Gravel fertilizer	4.9 ^{ab} (0.5)	0.47 (0.06)	11	38	5	36	57	4
Gravel soil	5.3 ^b (0.4)	0.53 (0.06)	7	44	7	15	49	5
PK	0.9 ^{ac} (0.3)	0.2 ^{ab} (0.21)	0	5	0	0	9	0
PK fertilizer	1.2 ^a (0.2)	0.08 ^a (0.06)	0	10	0	0	19	0
PK soil	4.5 ^b (0.3)	0.39 ^b (0.06)	1	33	18	0	64	0
PK sludge	0.4 ^c (0.2)	0.00 ^a (0.00)	0	2	0	0	6	0
Till	2.7 ^a (0.4)	0.32 (0.11)	1	17	7	11	28	0
Till fertilizer	4.4 ^b (0.5)	0.34 (0.05)	16	35	9	9	69	0
Till soil	4.9 ^b (0.4)	0.43 (0.06)	7	42	10	3	57	4
Till sewage	3.9 ^b (0.5)	0.39 (0.06)	33	61	4	41	72	0
50:50	3.3 ^a (0.3)	0.35 (0.06)	9	35	0	1	75	0
50:50 fertilizer	3.8 ^a (0.3)	0.35 (0.06)	6	31	6	10	61	1
50:50 soil	5.2 ^b (0.4)	0.38 (0.05)	5	46	5	24	56	0
50:50 sewage	4.0 ^a (0.3)	0.33 (0.06)	41	52	6	9	75	0
25:75	2.4 ^a (0.3)	0.23 ^a (0.09)	0	24	4	0	44	0
25:75 fertilizer	3.1 ^a (0.3)	0.12 ^b (0.06)	0	29	11	6	51	0
25:75 soil	5.6 ^b (0.5)	0.38 ^a (0.07)	0	52	11	11	74	0
25:75 sludge	3.2 ^a (0.5)	0.32 ^a (0.08)	2	33	4	7	30	0

PK, processed kimberlite.

Numbers in parentheses are standard error of means. Community evenness (E) is based on Simpson's Diversity Index (D) and calculated as $E = 1 - D$. Within substrates (gravel, PK, till, 50:50, 25:75), means which do not share a common letter are significantly different at $p = 0.050$.

litter may insulate soil, reducing depth and length of freeze (Hogg & Liefers 1991). Cargill and Chapin (1987) found high litter inhibited tundra plant germination, particularly shrubs, reducing nutrient benefits. The species pool was similar to other successful treatments, although vegetation was dominated by a few species (*Puccinellia nuttalliana*, *Epilobium angustifolium*), which may affect long-term diversity and resilience.

Fertilizer added less nitrogen and phosphorus to substrates than sewage and litter was less. Most significant effects of fertilizer on nitrogen were in till and 25:75, where finer textures and higher soil water improved nutrient retention and availability. Exceptional performance of till-fertilizer cannot be explained by nutrients, but by finer texture, more soil water and lower pH. Soil ameliorated pH and electrical conductivity but did not supply nutrients relative to sewage or fertilizer; it increased diversity and production of seed heads and flowers. Heavier application may extend benefits as observed by Mackenzie and Naeth (2010), who found increased organic carbon and nitrogen in 20 versus 10 cm. Soil may be of greater benefit for thermal regime as even thin organic layers maintain permafrost depth and surface temperature (Shirazi et al. 1998).

Our research supports others suggesting that without propagules and organic matter addition, recovery in the arctic will not occur on highly disturbed xeric sites (Cargill & Chapin 1987; Forbes et al. 2001). *Epilobium angustifolium* dominated natural recovery and is a common early successional species in the north. It often dominates following disturbance due to small, wind dispersed seed and low nutrient requirements, then dies and facilitates establishment of others (Bliss 1962; Viereck 1966; Baig 1992). Nitrogen and phosphorus were extremely high in some treatments relative to reference conditions,

potentially favoring graminoids and preventing natural recovery. Over time, seeded grasses spread, with similar abundance to natural recovery. Chapin and Shaver (1981) found *Calamagrostis canadensis* and *Arctagrostis latifolia*, used in our study, were important colonizers where organic matter was removed. Introducing various plant functional groups will increase community diversity and stability more than introducing more species (Forbes et al. 2001) as various growth forms may buffer environmental fluxes and low disturbance (Chapin et al. 1996). Ericaceous shrubs, *Betula glandulosa* and *Salix* dominate tundra and many propagate vegetatively. Ericaceous and deciduous shrubs were sensitive to disturbance and even after low impact disturbance did not regenerate (Chapin & Shaver 1981; Felix et al. 1992; Forbes et al. 2001). In oil sands reclamation, topsoil adds native propagules, including shrubs and forbs, accelerating development (Mackenzie & Naeth 2010). Soil was not a source of propagules in this study. Soil horizon admixing reduces the seed bank, and stockpiling even 6 months reduces propagule viability.

Forbes et al. (2001) suggest assessing restoration solely on vascular plants may be misleading as tundra resilience and stability relies on non-vascular plants. While vascular plants did not progress, moss colonized gravel in greater abundance than most treatments. Moss and lichen are important components of tundra biological crusts, regulating surface temperature and water (Van der Wal et al. 2001). Moss and lichen crusts may be most important early in succession with benefits in surface layers for seed and seedlings (Gold 1998) and may directly facilitate later communities (Bowker 2007). Fertilizer significantly increased biological crusts on reclamation sites in the Finnish arctic (Gretarsdottir et al. 2004) and moss was more abundant

in gravel-fertilizer and till-fertilizer. Repeated fertilizer may be required for full benefits.

Implications for Practice

- Improving reclamation substrates with fine materials and organic matter, combined with seeding, accelerates natural processes within 5 years facilitating initial tundra plant community development.
- Sewage sludge, readily available in mining camps, is as good as salvaged soil for initial plant community development in the arctic and should be used to supplement soil on large, severe disturbances.
- Target plant species and groups, particularly shrubs, must be added to restoration sites; natural recovery will not occur quickly and salvaged soil is not a good source of propagules.
- Surface added amendments eroded with time, exposing inhospitable substrates to germinating seed and roots and incorporation should be investigated.
- Even in human constructed harsh environments early ecosystem development is dynamic. Impacts of reclamation methods on soil and plant community development were not seen for 2–3 years and changed by year 5.

Acknowledgements

Diavik Diamond Mine Inc. is acknowledged for research funds and assistance in establishment and maintenance of research sites, and Arctic Alpine Seed Ltd. for native seed.

LITERATURE CITED

- Baig, M. N. 1992. Natural revegetation of coal mine spoils in the Rocky Mountains of Alberta and its significance for species selection in land restoration. *Mountain Research and Development* **12**:285–300.
- Billings, W. D. 1987. Constraints to plant growth, reproduction, and establishment in arctic environments. *Arctic and Alpine Research* **19**:357–365.
- Bliss, L. C. 1962. Adaptations of arctic and alpine plants to environmental conditions. *Arctic* **15**:117–144.
- Bowker, M. A. 2007. Biological soil crust rehabilitation in theory and practice: an underexploited opportunity. *Restoration Ecology* **15**:13–23.
- Bradshaw, A. 1997. Restoration of mined lands-using natural processes. *Ecological Engineering* **4**:255–269.
- Bruno, J. F., J. J. Stachowicz, and M. D. Bertness. 2003. Inclusion of facilitation into ecological theory. *Trends in Ecology and Evolution* **18**:119–125.
- Cargill, S., and F. S. Chapin III. 1987. Application of successional theory to tundra restoration: a review. *Arctic and Alpine Research* **19**:366–372.
- Carter, M. R. 1993. Soil sampling and methods of analysis. Lewis Publisher, Boca Raton, Florida.
- Chapin, F. S. III. 1983. Direct and indirect effects of temperature on arctic plants. *Polar Biology* **2**:47–52.
- Chapin, F. S. III, and G. R. Shaver. 1981. Changes in soil properties and vegetation following disturbance of Alaskan arctic tundra. *Journal of Applied Ecology* **18**:605–617.
- Chapin, F. S. III, M. Sydnora Bret-Harte, S. E. Hobbie, and H. Zhong. 1996. Plant functional types as predictors of transient responses of arctic vegetation to global change. *Journal of Vegetation Science* **7**:347–358.
- Densmore, R. V. 1994. Succession on regarded placer mine spoil in Alaska, U.S.A. in relation to initial site characteristics. *Arctic and Alpine Research* **26**:354–363.
- Drozdowski, B. L., M. A. Naeth, and S. R. Wilkinson. 2012. Evaluation of substrate and amendment materials for soil reclamation at a diamond mine in the Northwest Territories, Canada. *Canadian Journal of Soil Science* **92**:77–88.
- Felix, N. A., M. K. Reynolds, J. C. Jorgenson, and K. E. DuBois. 1992. Resistance and resilience of tundra plant communities to disturbance by winter seismic vehicles. *Arctic and Alpine Research* **24**:69–77.
- Forbes, B. C., and R. L. Jefferies. 1999. Revegetation of disturbed arctic sites: constraints and applications. *Biological Conservation* **88**:15–24.
- Forbes, B. C., J. J. Ebersole, and B. Strandburg. 2001. Anthropogenic disturbance and patch dynamics in circumpolar arctic ecosystems. *Conservation Biology* **15**:954–969.
- Gold, W. G. 1998. The influence of cryptogamic crusts on the thermal environment and temperature relations of plants in a high arctic polar desert, Devon Island, N.W.T., Canada. *Arctic and Alpine Research* **30**:108–120.
- Grettarsdottir, J., A. L. Aradottir, V. Vandvik, E. Heegaard, and H. J. B. Birks. 2004. Long term effects of reclamation treatments on plant succession in Iceland. *Restoration Ecology* **12**:268–278.
- Haag, R. W., and L. C. Bliss. 1973. Energy budget changes following surface disturbance to upland tundra. *Journal of Applied Ecology* **11**:355–374.
- Hall, S. L., C. D. Barton, and C. C. Baskin. 2010. Topsoil seedbank of an oak-hickory forest in eastern Kentucky as a restoration tool on surface mines. *Restoration Ecology* **18**:834–842.
- Harper, K. A., and G. P. Kershaw. 1997. Soil characteristics of 48-year-old borrow pits and vehicle tracks in shrub tundra along the CANOL No. 1 pipeline corridor, Northwest Territories, Canada. *Arctic and Alpine Research* **29**:105–111.
- Hogg, E., and V. Lieffers. 1991. The impact of *Calamagrostis canadensis* on soil thermal regimes after logging in northern Alberta. *Canadian Journal of Forest Research* **21**:387–394.
- Johnson, L. 1987. Management of northern gravel sites for successful reclamation. *Arctic and Alpine Research* **19**:530–536.
- Kidd, J. G., B. Streever, and M. Torre Jorgenson. 2006. Site characteristics and plant community development following partial gravel removal in an arctic oilfield. *Arctic Antarctic and Alpine Research* **38**:384–393.
- Li, X., and M. Y. P. Fung. 1998. Creating soil-like materials for plant growth using tailings sand and fine tails. *Journal of Canadian Petroleum Technology* **37**:44–47.
- Mackenzie, D. D., and M. A. Naeth. 2010. The role of the forest soil propagule bank in assisted natural recovery after oil sands mining. *Restoration Ecology* **18**:418–427.
- Magurran, A. E. 2004. Measuring biological diversity. Blackwell Publishing, Oxford, United Kingdom.
- Martínez-Ruiz, C., and B. Fernández-Santos. 2005. Natural revegetation on topsoiled mining-spoils according to the exposure. *Acta Oecologica* **28**:231–238.
- Peterson, K. M., and W. D. Billings. 1980. Vegetational patterns and succession in relation to microtopography near Atkasook, Alaska. *Arctic and Alpine Research* **12**:473–482.
- Prach, K., and P. Pyšek. 2001. Using spontaneous succession for restoration of human-disturbed habitats: experience from central Europe. *Ecological Engineering* **17**:55–62.
- Qian, P., J. J. Schoenauer, and R. E. Karamanos. 1994. Simultaneous extraction of available phosphorus and potassium with a new soil test: a modification of Kelowna extraction. *Communications in Soil Science and Plant Analysis* **25**:627–635.
- Reid, N. B., and M. A. Naeth. 2005. Establishment of a vegetation cover on kimberlite mine tailings: a field study. *Restoration Ecology* **13**:593–600.
- Shirazi, M. A., P. K. Haggerty, C. W. Hendricks, and M. Reporter. 1998. The role of thermal regime in tundra plant community restoration. *Restoration Ecology* **6**:111–117.
- Smyth, C. R. 1997. Early succession patterns with a native species mix on amended and unamended coal mine spoil in the Rocky Mountains of southeastern British Columbia, Canada. *Arctic and Alpine Research* **29**:184–195.

- Sokal, R. R., and F. J. Rohlf. 1995. *Biometry: the principles and practices of statistics in biological research*. W.H. Freeman and Company, New York.
- Truett, J. C., and K. Kertell. 1992. Tundra disturbance and ecosystem production: implications for impact assessment. *Environmental Management* **16**:485–494.
- Turner, M. G., W. L. Baker, C. Peterson, and R. K. Peet. 1998. Factors influencing succession: lessons from large, infrequent natural disturbances. *Ecosystems* **1**:511–523.
- Van der Wal, R., S. M. J. vanLieshout, and M. J. J. E. Loonen. 2001. Herbivore impact on moss depth, soil temperature and plant growth. *Polar Biology* **24**:29–32.
- Van Rensburg, L., M. S. Maboeta, and T. L. Morgenthal. 2004. Rehabilitation of co-disposed diamond tailings: growth medium rectification procedures and indigenous grass establishment. *Water, Air, and Soil Pollution* **154**:101–111.
- Vavrek, M. C., N. Fetcher, J. B. McGraw, G. R. Shaver, F. S. Chapin III., and B. Bovard. 1999. Recovery of productivity and species diversity in tussock tundra following disturbance. *Arctic Antarctic and Alpine Research* **31**:254–258.
- Viereck, L. A. 1966. Plant succession and soil development on gravel outwash of the Muldrow glacier, Alaska. *Ecological Monographs* **36**:181–199.
- Wong, M. H. 2003. Ecological restoration of mine degraded soils with emphasis on metal contaminated soils. *Chemosphere* **50**:775–780.